

ZENTRA / ZION EVENTS PLACE

Literal System Architecture

Repository-verified current-state structure and strict runtime flows

ARCHITECTURE VERDICT

Modular monolith with managed cloud services and event-driven integrations.

One Next.js deployment contains the public UI, admin UI, API handlers, server actions, authentication, domain services, and integration adapters. Supabase provides the application database and object storage. Render-hosted n8n, Gmail, and Gemini remain external integrations. This is not a microservice architecture.

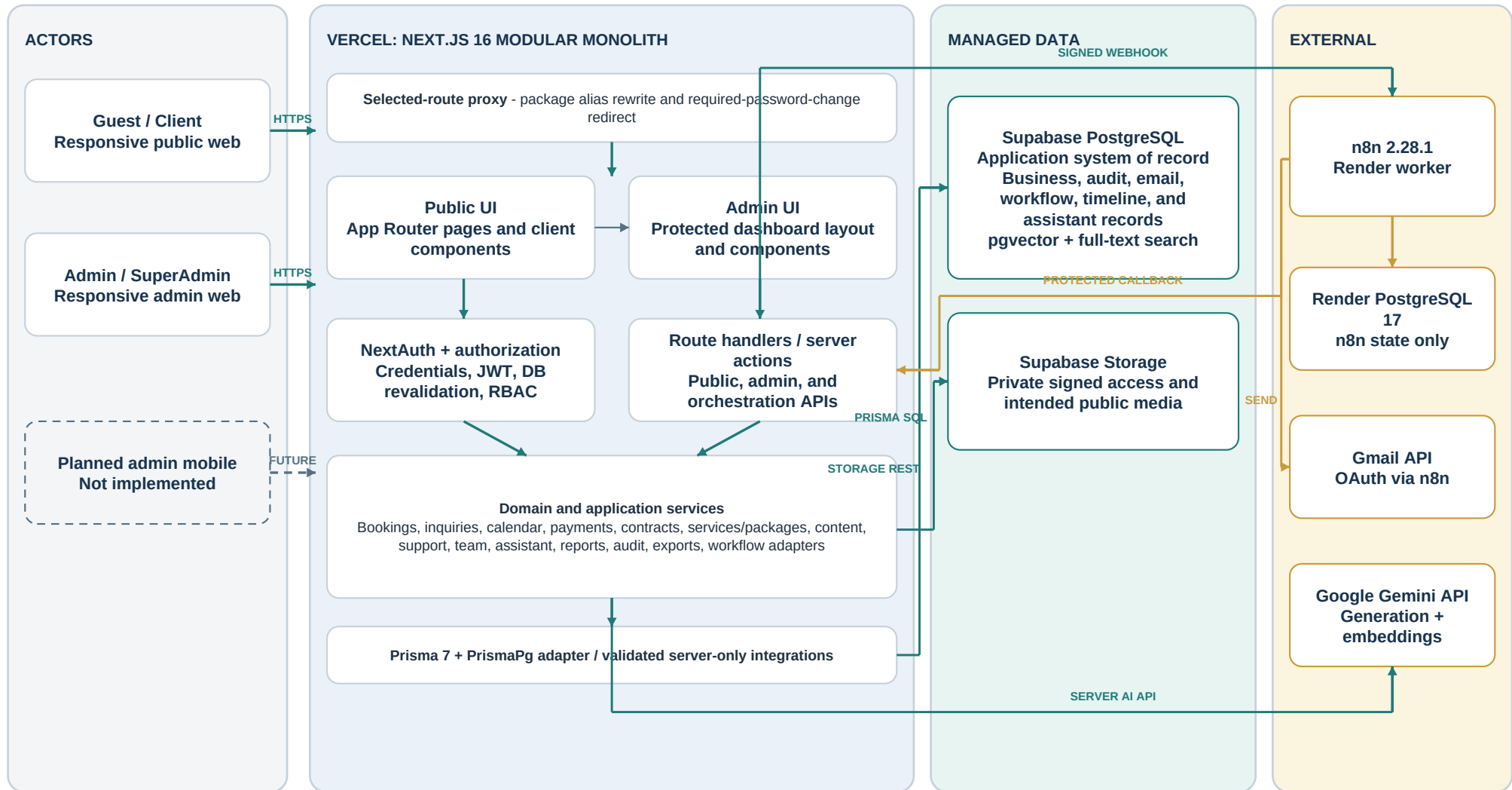
VERIFIED SCOPE

211 API route handlers
40 application pages
56 Prisma data models
30 database migrations

Evidence cut-off: **16 July 2026**

Literal logical architecture

Solid connections are implemented. Dashed connections are explicitly planned and are not current runtime paths.

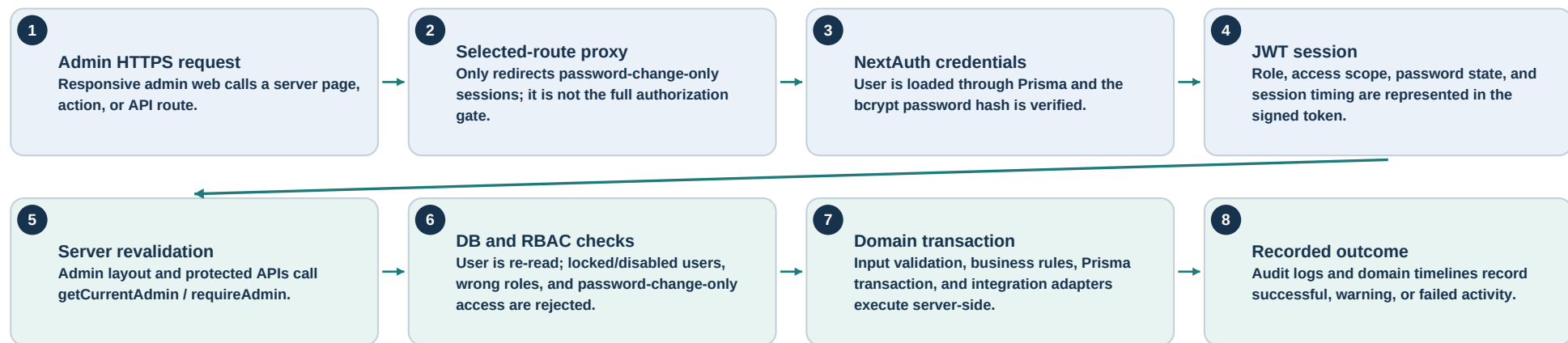


Strict boundary: browsers do not receive database credentials, storage service-role keys, Gemini keys, or n8n shared secrets.

Dashed mobile path is a constraint, not an implemented component.

Strict flow 1 - identity and protected administration

Every protected operation is authorized on the server. A client-side screen is never the security boundary.



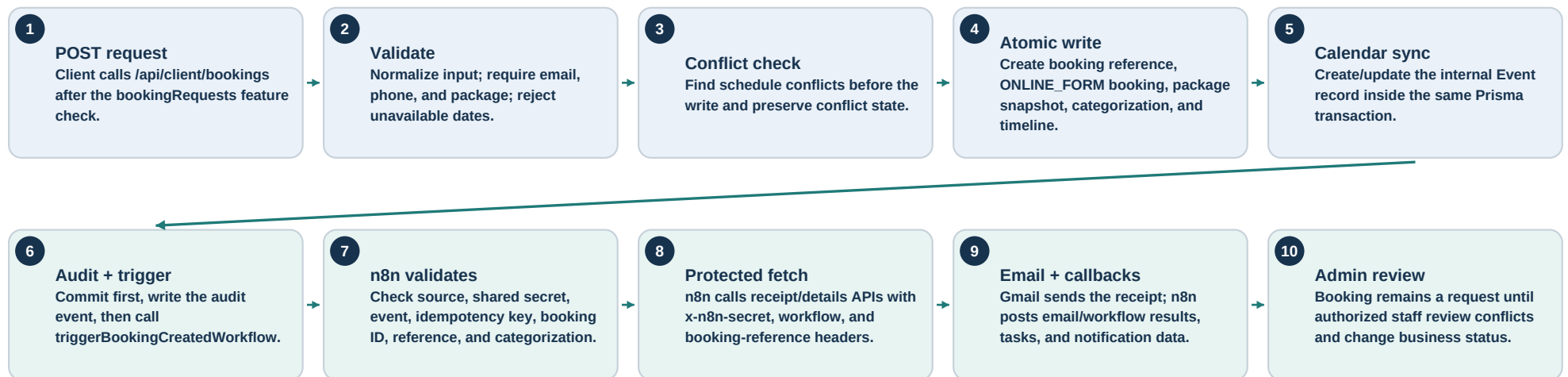
SuperAdmin rule: services/packages administration and other high-risk mutations use `requireSuperAdmin` or a SuperAdmin-only server layout. Admin and SuperAdmin are the only NextAuth administrative roles.

Client private-record access This is separate from admin NextAuth.



Strict flow 2 - client booking to operational follow-through

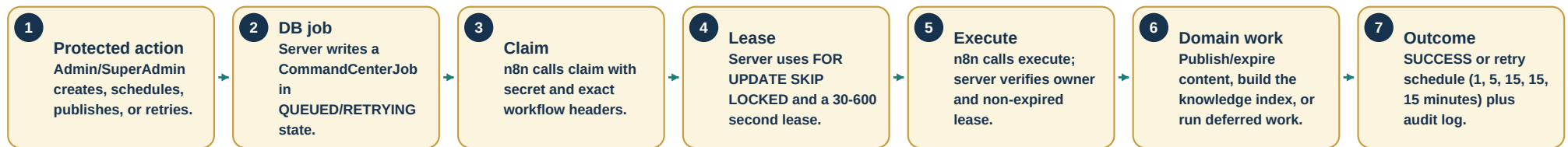
The database commit is authoritative. Automation follows the committed booking; it does not create the source booking record.



STATE OR FAILURE	LITERAL SYSTEM BEHAVIOR
Conflict found	Booking is still stored; syncStatus becomes CONFLICT_DETECTED and a timeline entry is created.
Webhook disabled or missing	The booking remains committed. The workflow log records CANCELLED or FAILED; no false email success is recorded.
n8n callback	Server endpoints validate shared secret/API key, source/workflow headers, booking reference, and rate limits before writing results.
Manual admin booking	Uses requireAdmin, the same conflict/domain transaction path, and audit logging; it does not call the client booking-created webhook.

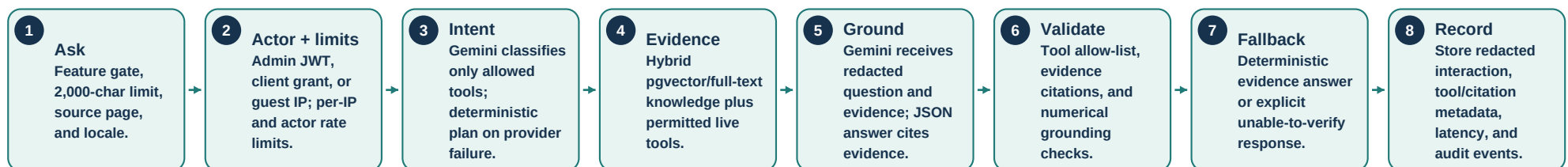
Strict flow 3 - Command Center worker and grounded assistant

A. Command Center job execution n8n coordinates work through protected APIs; it does not directly modify the application database.



Data ownership: Supabase PostgreSQL stores jobs and business state. Render PostgreSQL stores only n8n internal state. Expired leases are reclaimable; exhausted attempts become FAILED until an authorized retry.

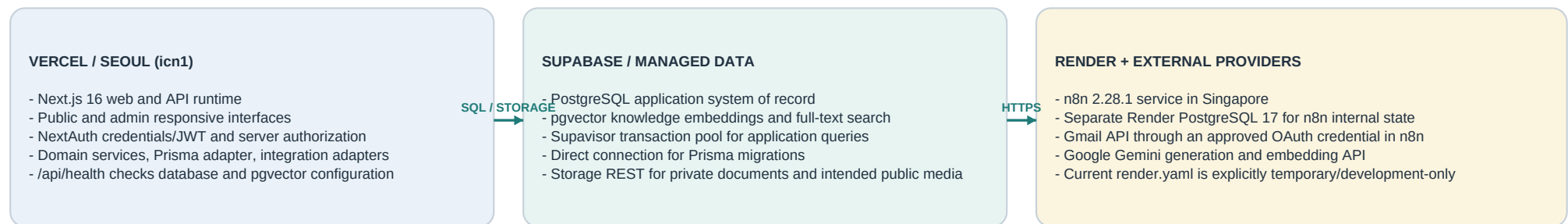
B. Smart Assistant answer path Answers are grounded in approved knowledge and live records, with deterministic fallbacks.



Assistant privacy boundary: public users can access public tools only. Verified clients can access their own booking-related tools. Admins are still restricted by the defined tool set. Gemini API keys and raw database access remain server-side.

Deployment topology, ownership, and non-negotiable boundaries

The topology below is configured by repository deployment files; it is not a claim that every component is currently live or production-ready.



BOUNDARY	AUTHORITATIVE OWNER	ALLOWED ACCESS	DISALLOWED PATH
Application records	Supabase PostgreSQL	Next.js server through PrismaPg; migration CLI through DIRECT_URL.	No direct browser or n8n database credentials.
Object files	Supabase Storage	Server-only REST adapter; signed URLs for private access and public URLs only where intended.	No service-role key in browser or NEXT_PUBLIC variables.
Workflow state	Render PostgreSQL	n8n internal execution/configuration state only.	Must not become a second application system of record.
Email delivery	Gmail through n8n	Backend-prepared payload; n8n sends and reports result through protected callbacks.	No browser-held OAuth credential or unverified success state.
AI processing	Gemini API + app controls	Server sends redacted question and selected evidence; embeddings stored in pgvector.	No direct client key and no unrestricted model access to the database.
Operational evidence	Application database	Audit logs, timelines, workflow logs, email logs, assistant interaction logs, and health route.	Do not treat UI notifications alone as proof of completion.

Final architecture verdict and strict implementation rules

Use the system as a modular monolith.
Keep one deployable Next.js application while preserving module boundaries, transactions, server authorization, and integration adapters. n8n is an external orchestrator, not a business-data service. Supabase PostgreSQL remains the single application system of record. This structure matches the code that exists today and avoids inventing microservices that the repository does not contain.

1. Single entry boundary
All browser and future mobile traffic must use HTTPS pages/APIs. No client may connect directly to PostgreSQL, service-role storage, n8n secrets, or Gemini keys.

2. Server authorization
Protected pages and every protected mutation must call requireAdmin or requireSuperAdmin. The proxy is only an early redirect/rewrite layer.

3. Transactional business state
Booking, payment, contract, content, and job state changes must pass through domain services and Prisma transactions where atomicity is required.

4. Traceable effects
Every material administrative or automated effect must leave authoritative audit, timeline, email, workflow, or assistant-interaction evidence.

5. Secret orchestration
n8n requests must use the configured shared secret/API key, exact source/workflow headers, reference checks, rate limits, and idempotency or lease controls.

6. Mobile parity
A future admin mobile application must call the same secured APIs and role rules. Mobile-only direct data paths would create a second, unsafe architecture.

Current-state constraints that the diagram does not conceal

STATUS	VERIFIED STATEMENT
Implemented	Responsive public and admin web interfaces; modular domain/API/data layers; server-side RBAC; Supabase database/storage adapters; n8n and Gemini integration code.
Not implemented	Native admin mobile application, Android/iOS project, service worker, or web-app manifest/PWA packaging.
Not production evidence	The Render blueprint is marked temporary/development-only and uses free services. Repository deployment configuration does not prove that all hosted services are active.
Current readiness gap	Production environment validation previously identified an invalid public auth URL and missing client-access webhook/malware-scanner configuration. Resolve and re-run verification before release.